

2. Agresti - Coull CI.

$$n \geq 30$$

04/24 F ①

Idea: Adjust the data before constructing the confidence interval.

Adjusted values: Add 2 successes and 2 failures in the data.
add 4 items in n .

[Advantages: works well for small or moderate samples]

After adjustment, do Wald CI.

$$\tilde{n} = n + 4$$

$$\tilde{x} = x + 2$$

← ^{adjusted.} # of successes in the sample.

$$\tilde{p} = \frac{\tilde{x}}{\tilde{n}}$$

Then $(1 - \alpha) 100\%$ Agresti - Coull CI. is.

$$\tilde{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}}$$

b) 95% Agresti - Coull CI.

$$\tilde{p} = \frac{22 + 2}{100 + 4} = 0.2308$$

$$\begin{aligned} \tilde{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}} &= 0.2308 \pm 1.96 \sqrt{\frac{0.2308(1 - 0.2308)}{100 + 4}} \\ &= (0.1498, 0.3117) \end{aligned}$$

3. Score test (Z-test)

(2)

$$\text{Test: } H_0: p = p_0 \quad \text{vs.} \quad \begin{cases} H_a: p \neq p_0 & \text{two-sided} \\ H_a: p > p_0 & \text{right-sided} \\ H_a: p < p_0 & \text{left-sided} \end{cases}$$

1) 3-step test.

① ~~test~~ test statistic.

$$Z^* = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

② critical value.

$$\text{two-sided} = Z_{\frac{\alpha}{2}}$$

$$\text{right-sided} = Z_{\alpha}$$

$$\text{left-sided} = -Z_{\alpha}$$

③ Decision and conclusion.

$$\text{two-sided: If } |Z^*| > Z_{\frac{\alpha}{2}}, \text{ then reject } H_0$$

$$\text{right-sided: If } Z^* > Z_{\alpha}, \text{ then reject } H_0$$

$$\text{left-sided: If } Z^* < -Z_{\alpha}, \text{ then reject } H_0$$

2) p-value

$$\text{two-sided: p-value} = 2P(Z > |Z^*|)$$

$$\text{right-sided: p-value} = P(Z > Z^*)$$

$$\text{left-sided: p-value} = P(Z < Z^*)$$

If p-value $< \alpha$, then reject H_0